

Reciprocal Rank Fusion and Hybrid Retrieval

Hybrid Retrieval Motivation

Neither vector search nor keyword search is optimal for all query types. Sparse BM25 excels at exact keyword matching (part numbers, technical terms), while dense vector search captures conceptual semantics.

RRF Formula & Algorithm

Reciprocal Rank Fusion (RRF) combines ranked lists from multiple independent retrievers without requiring score normalization. RRF calculates a final score for each document using the sum of reciprocal ranks:

$RRF_score(d) = \sum (1 / (k + r_i(d)))$, where k is a constant parameter (typically 60) and $r_i(d)$ is document d 's rank in retriever i .

Evaluation & Benefits

RRF improves Mean Reciprocal Rank (MRR) and NDCG@10 by up to 18% over single-retriever baselines across diverse benchmarks.